# KINETIC THEORY

**1.** In dynamic equilibrium, molecules collide & change their speed during collision. Only ____ properties are constant.
(1) instantaneous (2) average
(3) both (1) & (2) (4) NOT

**2.** Match the following :
(a) PV = constant (1) Charle's Law
(b) $V \propto T$ (2) GayLussac's Law
(c) $P \propto T$ (3) Boyle's Law
(1) a – 1, b – 3, c – 2
(2) a – 3, b – 2, c – 1
(3) a – 3, b – 1, c – 2
(4) a – 1, b – 2, c – 3

**3.** Experimental value of specific heat of gases like $CH_4$, $C_2H_6$ (polyatomic gases) are :-
(1) less than predicated value
(2) more than predicted value
(3) equal to predicated value
(4) Anything is possible

**4.** When gases diffuse, then rate of diffusion is :-
(1) m (2) $m^{3/2}$ (3) $m^{1/2}$ (4) $m^{-1/2}$

**5.** The mean free path in gases, is of the order____.
(1) thousands of angstroms
(2) thousands of micrometers
(3) thousands of pico meters
(4) thousands of fermimeters

**6.** Internal energy of an ideal gas depends on :-
(1) Temperature
(2) Volume
(3) Pressure
(4) depends on type of process

**7.** Why total KE & momentum conserved for ideal gas ?
(1) all collisions between molecules & even wall are elastic
(2) $F_{net} = 0$
(3) all collision are in-elastic
(4) None of these

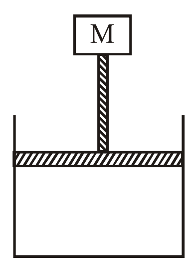
**8.** How is viscosity, heat conductivity & diffusion of gas (bulk properties) estimated ?
(1) Force
(2) Momentum
(3) Molecular size
(4) Weight of gas sample

**9.** If the pressure in a closed vessel is reduced by drawing out some of the gas, the mean free path:
(1) is increased
(2) is decreased
(3) remains unchanged
(4) increases or decreases according to the nature of the gas

**10.** Hydrogen and nitrogen are at same temperature. The molecules of which one of them have more kinetic energy :
(1) Hydrogen
(2) Nitrogen
(3) Both have equal amount of energy
(4) Depends upon actual value of temperature

**11.** Pressure of an ideal gas is increased by keeping temperature constant. What is its effect on kinetic energy of molecules.
(1) Increases
(2) Decreases
(3) No change
(4) Cannot be determined

**12.** A cylinder containing an ideal gas has a piston of mass M is in vertical position and that is able to move up or down without friction as shown in figure. If the temperature is increased.




(1) Both P and V of the gas will change
(2) Only P will increase according to charle's law
(3) V will change but not P
(4) P will change but not V

**13.** The pressure of a gas is decided by :
(1) random collisions on unit area
(2) average collisions per unit time on unit area of wall
(3) average collisions on unit area
(4) random collisions per unit time on unit area

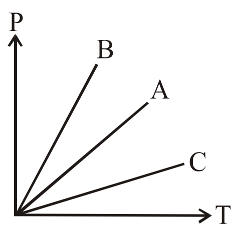
**14.** Which of the quantity is zero on an average for molecules of an ideal gas in equilibrium.
(1) Kinetic energy (2) Momentum
(3) Density (4) Speed

**15.** The root mean square speed of the ideal gas molecule is doubled if.
(1) Pressure is doubled at constant volume
(2) Pressure is made four times at constant volume
(3) Volume is doubled at constant pressure
(4) Volume is increased by about 42% at constant pressure

**16.** Specific heat capacity of a solid is
(1) R (2) 2R (3) 3R (4) 4R

**17.** Specific heat capacity of water is :-
(1) 3R (2) 6R (3) 5R (4) 9R

**18.** Pressure verses temperature graph of an ideal gas at constant volume is shown by a straight line-A in figure. If the mass of the gas is doubled and volume is halved, the corresponding pressure verses temperature graph is givey by line :




(1) A (2) B (3) C (4) None

**19.** Molecules of air in a room do not fall and settle on the ground because of:-
(1) all molecules having high temperature
(2) their high speed and succesive collision
(3) they collide perfectly inelastically
(4) None of the above

**20.** Gas leaking from a cylinder in a kitchen takes considerable time to diffuse to the other corners of the room why :
(1) molecules of gas have rather slow speed than sound
(2) They collide perfectly inelastically and stick to each other
(3) Due to collision of molecules they can not move straight unhindered
(4) None of these

**21.** According to the kinetic theory of gas, the temperature of a gas is a measure of average:-
(1) Velocity of it's molecules
(2) Linear momenta of it's molecules
(3) Kinetic energy of it's molecules
(4) Angular momenta of it's molecules

**22.** Law of equipartition of energy is used to predict:-
(1) Specific heat of gases
(2) Specific heat of solids
(3) both (1) & (2)
(4) Neither (1) nor (2)

**23.** When an ideal gas is compressed adiabatically, it's temperature rises, the molecules on the average have more kinetic energy than before. The kinetic energy increases because of:-
(1) Collisions with moving parts of wall only
(2) Collision with the entire wall
(3) molecules gets accelerated in their motion inside the volume
(4) redistribution of energy among the molecules

**24.** The internal energy of an ideal gas is in the form of:-
(1) Kinetic energy of molecules
(2) Potential energy of molecules
(3) Both kinetic and potential energy of molecules
(4) Gravitational potential energy of molecules

**25.** As temperature tends to zero i.e $T \rightarrow 0$ K
(1) Specific heat of all substance approaches zero
(2) Specific heat of all substance aproaches infinity
(3) Specific heat of substance may be zero or infinity
(4) None of the above

**26.** Some gas at 300K is enclosed in a container now, the container is placed on a fast moving train. While the train as in motion, the temperature of the gas.
(1) Rise above 300K (2) Falls below 300K
(3) Remains unchanged (4)Becomes unsteady

**27.** Choose the correct option.
(1) $C_P – C_V = R$ is true for monoatomic ideal gas
(2) $C_P – C_V = R$ is true for poly atomic ideal gas
(3) $C_P – C_V = R$ is true for any type of ideal gas
(4) None of the above